

PORONAYSK, Russian Federation

WMO: 320980

Lat: 49.217N

Lon: 143.100E

Elev: 26

StdP: 14.68

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-14.9	-24.4	1.4	-18.3	-20.5	1.8	-13.6	25.1	21.5	22.8	19.5	4.7	320	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	72.4	64.3	69.0	63.3	66.4	62.4	66.6	70.1	64.8	67.7	63.1	65.6	8.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	93.9	68.1	63.5	87.9	66.3	62.0	83.0	64.8	31.4	69.8	29.9	67.9	28.7	65.8	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.2	15.9	-27.3	82.8	4.7	3.7	-30.7	85.5	-33.5	87.6	-36.1	89.7	-39.6	92.3		
(5)			-27.5	69.8	4.7	2.8	-30.9	71.8	-33.7	73.5	-36.3	75.1	-39.7	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	4.9	7.1	19.0	32.0	41.2	50.1	57.4	60.8	54.0	41.2	24.8	8.8		
	DBStd	20.82	10.57	9.49	8.09	5.11	5.53	5.19	4.44	4.46	5.60	6.20	9.83	10.32		
	HDD50	6774	1397	1201	960	540	280	58	1	0	24	278	757	1278		
	HDD65	11483	1862	1621	1425	990	737	447	239	144	331	737	1207	1743		
	CDD50	788	0	0	0	0	8	62	232	336	144	6	0	0		
	CDD65	21	0	0	0	0	0	1	4	15	1	0	0	0		
	CDH74	81	0	0	0	0	1	12	18	42	8	0	0	0		
	CDH80	13	0	0	0	0	0	2	4	6	1	0	0	0		
	WSAvg	7.1	7.5	6.9	7.3	7.5	7.1	6.4	6.2	6.7	7.1	7.7	7.4	7.2		
	PrecAvg	29.5	0.9	1.0	1.5	2.1	2.7	2.3	3.1	3.7	4.7	3.7	2.0	1.5		
(15) Precipitation	PrecMax	45.4	3.3	4.2	4.4	5.2	7.2	5.9	7.4	14.5	10.7	8.7	5.0	4.8		
	PrecMin	17.6	0.0	0.0	0.2	0.5	0.4	0.2	0.2	0.3	0.9	0.6	0.0	0.3		
	PrecStd	6.5	0.7	0.9	1.0	1.2	1.5	1.4	1.7	3.0	2.4	1.9	1.4	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	32.5	38.2	52.7	65.2	74.4	75.3	78.8	73.7	61.3	49.5	35.0		
		MCWB	29.8	29.4	32.7	41.6	51.4	60.6	67.4	66.7	61.1	50.8	43.9	33.4		
	2%	DB	28.9	28.7	34.8	45.1	57.8	65.1	69.4	73.3	68.6	57.1	44.7	32.2		
		MCWB	26.8	26.5	30.7	38.8	49.4	58.6	64.1	65.8	60.1	50.3	41.4	30.5		
	5%	DB	26.0	25.8	32.4	41.5	52.9	61.0	65.7	70.0	65.5	54.4	41.8	29.0		
		MCWB	23.8	23.3	29.6	37.0	47.1	56.7	62.6	65.0	59.6	49.2	38.6	27.0		
	10%	DB	22.0	22.5	30.9	38.9	49.6	57.7	63.6	67.6	63.1	52.0	38.5	25.4		
		MCWB	19.8	20.2	28.6	35.4	45.5	54.4	61.2	64.0	58.6	47.7	35.1	23.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.5	30.2	34.0	43.0	53.3	62.8	67.9	70.0	65.1	55.2	45.4	34.0		
		MCDB	30.7	31.8	37.2	49.9	60.7	70.4	74.9	74.7	69.8	58.8	48.1	34.8		
	2%	WB	26.8	26.8	31.6	39.7	50.3	59.5	64.9	67.8	62.7	52.5	42.3	30.8		
		MCDB	28.5	28.8	33.3	43.7	55.9	64.0	68.1	70.9	65.4	55.0	44.3	32.1		
	5%	WB	23.7	23.4	30.3	37.7	47.8	56.9	63.0	66.1	61.1	50.7	38.9	27.2		
		MCDB	26.0	25.6	31.9	41.0	51.7	60.5	65.2	69.0	64.0	53.2	41.4	28.9		
	10%	WB	19.8	20.1	28.9	35.7	45.8	54.7	61.5	64.5	59.4	48.4	35.4	23.5		
		MCDB	21.9	22.3	30.9	38.4	49.2	57.5	63.4	67.1	62.3	51.2	38.0	25.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	19.2	15.9	9.9	10.3	9.8	7.7	9.7	14.9	15.2	14.2	16.1		
		MCDBR	10.8	14.7	14.3	14.8	18.9	19.0	12.5	15.0	18.5	19.3	15.6	12.6		
		MCWBR	10.6	13.5	11.7	9.8	11.4	11.6	8.0	9.0	11.7	13.6	12.8	12.3		
		MCDBR	10.7	14.4	13.1	13.7	16.9	16.6	11.2	12.3	14.7	15.8	14.4	12.4		
	5% WB	MCWBR	10.6	13.3	11.4	9.6	11.0	10.6	7.5	8.2	10.7	13.3	13.2	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.324	0.392	0.481	0.520	0.486	0.455	0.428	0.385	0.393	0.351	0.305		
	taud		2.099	1.986	1.890	1.847	1.871	2.078	2.245	2.333	2.388	2.174	2.102	2.091		
	Ebn at Noon		232	250	251	241	238	249	255	257	258	225	203	205		
	Edh at Noon		27	39	51	60	61	50	42	36	31	32	27	24		
(44) All-Sky Solar Radiation	RadAvg		439	780	1129	1365	1519	1653	1430	1315	1116	714	406	318		
	RadStd		40	49	108	153	167	149	161	132	98	50	40	31		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page